Sri Ram Nulu

ee24btech11217@iith.ac.in | sriram.nulu@proton.me | +91 9100810329 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.13
FIITJEE JUNIOR COLLEGE	Class XII (BIEAP)	2024	94.40%
SIR C R REDDY PUBLIC SCHOOL	Class X (CBSE)	2022	95.80%

SKILLS

HW Description Languages : Verilog, SystemVerilog

Programming Languages : C, Python

Tools : AMD Vivado, iVerilog, GTKWave, Git, KiCad

Core Skills : RTL Design, RTL Verification

WORK EXPERIENCE

Undergraduate Researcher | ***AESICD Laboratory, IIT Hyderabad***

Under the supervision of Dr. Amit Acharyya (Professor, Dept. of EE)

Project: **CNN Hardware Accelerator for AI Inference**

- Developing a hardware-efficient YOLO-based Convolutional Neural Network inference engine in Verilog.
- Designed Non-linear Activation modules, and co-designed Systolic Array based Convolution modules for AI inference pipeline.

PROJECTS

Systolic Array based Depth-Wise & Point-Wise Convolution Modules

- Co-designed parametrizable output-stationary systolic array micro-architecture for Point-Wise Convolution and a dedicated Chunk Formatter to orchestrate 128 bits/cycle dataflow from external bus.
- Optimized legacy Depth-Wise convolution datapath to eliminate redundant blocks and enable direct interconnect, reducing latency by 40%.
- Built parametrizable Python framework for input chunk and golden-model generation and SystemVerilog testbenches. Verified Depth-Wise and Point-Wise Convolution designs for Systolic Array sizes 4x4 to 32x32 and 4x4 to 128x128 respectively.

Pipelined Non-linear Activation Function (GeLU & SiLU) Modules

- Designed 5-stage pipelined, synthesizable non-linear activation units (GeLU & SiLU) in FXP16 format in Verilog using piecewise-linear (PWL) approximation and non-uniform power-of-two segmentation program in C.
- Achieved absolute maximum errors of ~2, ~1 and ~1 LSBs in Q4.12, Q6.10 and Q8.8 formats.
- Delivered the design code for FPGA testing at 100 MHz. Design passed functional verification.

Pipelined RV32I Processor Core | GitHub

Personal Project

- Designed a synthesizable, 5-stage pipelined RISC-V Processor Core in SystemVerilog with supports for Data Hazards and Control Hazards, supporting all 37 base-integer instructions in the RV32I ISA.
- Automated verification using official RISC-V Tests Repository using custom Bash script

Simulation for Gate Current Calculation using Transfer Matrix Method | GitHub

Group Project under Coursework

- Designed a C program to compute the transmission probability T(E) of an electron in tunneling across the oxide (barrier) from the left (channel) to the right (gate) using the Transfer Matrix, given the electric potential profile across the barrier and the electron energy E.
- Validated the functionality using three distinct base potential profiles across an electron energy sweep.

RELEVANT COURSEWORK

Basic Electrical: Circuits & Network Analysis, Signals & Systems, Control Theory

VLSI: Computer Architecture, Digital Systems, Analog Electronics & Lab, Semiconductor Device Fundamentals, CMOS Processing, Numerical Methods for Device Modeling

Basic Engineering: Intro to Programming, Data Structures & App.

Mathematics: Probability, Calculus, Vector Calculus, Differential Equations, Matrix Theory

VOLUNTEERING & POSITION OF RESPONSIBILITIES

Volunteer | National Service Scheme (NSS)

- Participated in several Service Activities, Community Initiatives, Cultural Events and Awareness Programs.
- Contributed 100+ hours in activities like Swachh Bharat, Tree Protection, Talks, et cetera.

Student Representative | JEE Open Day 2026

- Represented the Department of Electrical Engineering at IIT Hyderabad, advising JEE Advanced 2026 qualifiers on curriculum, department culture, and campus life.